Cosmological Distances

From redshift to the age of the universe

A course at three levels — plain language, undergraduate, graduate

With the interactive program ***`redshift_distance_gui.py`***

August 2026 edition

(all numerical values re-checked with *SageMath*, see appendix B)

Planck 2018 cosmology — $H_0 = 67.66\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ $\quad \Omega_\mathrm{m} = 0.3111 \quad \Omega_\Lambda = 0.6889$

Purpose of this document. This course presents, in detail and with derivations, the physics that turns a *cosmological redshift* z into several notions of distance and time, within the standard ΛCDM model constrained by the *Planck 2018* data [1].

Each notion is presented at three levels:

Level 1 — Plain language	analogies, mental images, no equations
Level 2 — Undergraduate	calculus and integrals, no tensors
Level 3 — Graduate	general relativity, FLRW, complete derivations

Chapter 1 describes the graphical interface of `redshift_distance_gui.py` in detail and explains every displayed quantity, including the cryptic "TT,TE,EE+lowE+lensing+BAO" that names the data combination behind the cosmological parameters used.

All the numerical formulas are those implemented in the `astropy.cosmology` library (`Planck18` cosmology) [21].

Contents

How to read this course

This course is meant to be readable at several depths.

Level 1 — Plain language

To grasp only *the idea*, the blue boxes like this one are enough: no equations, but the overall picture. Everything else can be skipped.

Level 2 — Undergraduate

With an undergraduate background in physics or engineering, the green boxes add the essential formulas and short derivations using calculus and integrals only — no general relativity.

Level 3 (main text). For a reader familiar with general relativity (metric tensor, Einstein equations, index notation), the main text gives the rigorous derivations, from the FLRW metric down to the cosmological distances.

Going further

Orange "Going further" boxes: historical background, current debates, conceptual subtleties.

Worked example

Green "Worked example" boxes: numbers for emblematic redshifts.

In the program

Teal "In the program" boxes: what the quantity corresponds to in the companion application `redshift_distance_gui.py`.

Warning — conceptual trap

Red "Warning" boxes: conceptual traps that popular accounts spread.

Notation and conventions

Signature The metric has signature $(-,+,+,+)$.

Greek indices $\mu, \nu \in \{0,1,2,3\}$, spacetime indices.

Latin indices $i, j \in \{1,2,3\}$, spatial indices.

c — $299\,792.458\,\mathrm{km\,s^{-1}}$ (exact by definition since 1983). We do not use the convention $c = 1$.

H_0 — The Hubble constant today, in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$. Planck 2018 value: 67.66.

h — Reduced Hubble parameter, $h \equiv H_0/(100\,\mathrm{km\,s^{-1}\,Mpc^{-1}})$. Equal to 0.6766 in Planck 2018.

z — Cosmological redshift, dimensionless. Always positive for cosmological objects (the universe expands).

$a(t)$ — Scale factor, normalised so that $a(t_0) = 1$ today.

Mpc — The megaparsec, 1 Mpc $\approx 3.086 \times 10^{19}$ km $\approx 3.262 \times 10^{6}$ ly.

ly — The light-year, 1 ly $= 9.461 \times 10^{12}$ km.

Gly — The giga-light-year, 1 Gly $= 10^{9}$ ly.

Gyr — The gigayear (10^{9} years).

Remark 0.1. *Every distance introduced in this course is* model dependent*: outside the local universe ($z \ll 1$) there is no unique notion of distance. Turning a redshift into a distance always presupposes a model — here ΛCDM with the Planck 2018 parameters.*

Part I

Reading the program and the Planck 2018 parameters

Chapter 1

The graphical interface `redshift_distance_gui.py`

The program shipped with this course is a Qt6 application (PyQt6 + pyqtgraph) that displays, for a redshift z entered by the user, four cosmological distances and several derived quantities, together with a log-log plot of these distances against z. It is bilingual (French / English): the *Language* menu switches immediately.

1.1 Overview

The interface has four areas:

1. **Header**: title and the cosmological reference line.
2. **Redshift panel**: the z field, the preset buttons and the field that looks up an object by its name.
3. **Model panel**: spatial curvature Ω_k and the SH0ES comparison.
4. **Left panel**: four distances and the cosmological quantities; **right panel**: an interactive plot of the four distances against z (logarithmic axes), with a vertical cursor at the current z and markers for $z = 1$, the maximum of D_A, GN-z11 and the CMB.

1.2 The subtitle line

The line displayed under the title reads:

$$\textit{Planck 2018 (TT,TE,EE+lowE+lensing+BAO)} \text{ — } H_0 = 67.66\ \mathit{km/s/Mpc} \quad \Omega_\mathrm{m} + \Omega_\nu = 0.31110$$
$$\Omega_\Lambda = 0.68885 \quad \Omega_\gamma = 5.402 \times 10^{-5} \quad t_0 = 13.7869\ \mathit{Gyr}$$

It is not decorative: it states *exactly* which model of the universe the program uses. Let us decode it.

Level 1 — Plain language

In one sentence. "The program uses the parameters of the universe that the Planck space mission determined in 2018, by combining everything we know about the relic radiation (the CMB) plus one independent external data set."

1.2.1 “Planck 2018”

Planck is the European Space Agency satellite launched in 2009, which observed the cosmic microwave background (CMB) until 2013; its data were released progressively. The final release, “Planck 2018 results” (published in 2020 in *A&A* 641, A6), gives the most precise values to date of the parameters of the standard cosmological model [1].

1.2.2 $H_0 = 67.66$ km/s/Mpc — the Hubble constant

This is the *rate* at which the universe expands today. A galaxy at 1 Mpc from us (feeling no local force) recedes on average at 67.66 km/s. At 10 Mpc it recedes ten times faster (Hubble’s law).

Warning — conceptual trap

H_0 has the dimension of an *inverse time*: $H_0 \approx 2.19 \times 10^{-18}$ Hz. Its inverse, the *Hubble time*, is $1/H_0 \approx 14.5$ Gyr, which is also the order of magnitude of the age of the universe (13.79 Gyr) — and that is no coincidence.

1.2.3 $\Omega_\mathrm{m} = 0.3111$ — the total matter density

This is the fraction of the total energy of the universe, today, in the form of *matter* (meaning “massive non-relativistic particles”), of all kinds:

- $\Omega_b \approx 0.049$ ordinary *baryons* (atoms, stars, gas, planets, and so on);
- $\Omega_\mathrm{cdm} \approx 0.262$ *cold dark matter*: massive collisionless particles of unknown nature.

So only $\sim 5\%$ of everything that weighs in the universe is made of known matter; the remaining $\sim 26\%$ of matter is dark and detected only through gravity.

1.2.4 $\Omega_\Lambda = 0.6889$ — the cosmological constant (“dark energy”)

The rest of the total energy ($\sim 69\%$) takes the form of “dark energy”, attributed in the minimal model to a *cosmological constant* Λ (an extra term in the Einstein equations). It produces the **acceleration** of the expansion, observed for the first time in 1998 [16, 17].

1.2.5 Why is $\Omega_\mathrm{m} + \Omega_\Lambda \approx 1$?

Because space is **flat**. The sum of all energy densities (matter + radiation + Λ) must equal exactly 1 at zero curvature. Measurement gives $\Omega_\mathrm{m} + \Omega_\Lambda = 0.9999 \pm 0.002$, which justifies assuming flatness. See chapter 6, section 6.4.

1.2.6 “TT,TE,EE+lowE+lensing+BAO” — the cryptic phrase

This is the **list of data sets** combined by the Planck collaboration to constrain all parameters simultaneously. Each acronym names an observation:

Acronym	Full name	What it measures
TT	Temperature–Temperature	Angular power spectrum of the CMB in *intensity* (the temperature contrasts on the sky, $\Delta T/T \sim 10^{-5}$). The "historical" CMB measurement.
TE	Temperature–E-mode	Cross-correlation between temperature and E-mode polarisation (the curl-free component).
EE	E-mode–E-mode	The E-mode polarisation spectrum alone. Independent of TT.
lowE	low-ℓ E-mode polarisation	The EE spectrum at large angular scales ($\ell < 30$, i.e. angles $> 6°$). Very sensitive to the reionisation optical depth.
lensing	CMB lensing	Gravitational lensing of the CMB by large-scale structure. The intervening structure slightly distorts the CMB map; that distortion is extracted statistically.
BAO	Baryon Acoustic Oscillations	Acoustic oscillations observed *today* in the galaxy distribution (SDSS, BOSS, eBOSS). A peak at $\sim$ 150 Mpc comoving, a relic of the sound waves of the primordial universe. *External* to Planck, but combined with it.

Level 1 — Plain language

Why combine everything? A single observation constrains a few parameters but leaves others free; combining independent measurements breaks the degeneracies and yields a final precision much better than any single measurement. Concretely, the CMB (TT, TE, EE) mostly fixes the global geometry and the composition at recombination, while BAO calibrates the present-day *scale*. Together they constrain H_0, Ω_m and Ω_Λ to $\sim 0.5\%$.

Level 2 — Undergraduate

Angular power spectrum. The CMB temperature map on the celestial sphere is expanded in spherical harmonics:

$$\frac{\Delta T}{T}(\theta,\phi) = \sum_{\ell,m} a_{\ell m}\, Y_{\ell m}(\theta,\phi), \qquad C_\ell^{TT} = \frac{1}{2\ell+1}\sum_m |a_{\ell m}|^2.$$

ℓ is the *multipole* (the spherical analogue of a Fourier mode); $\ell \sim 200$ corresponds to an angle of $\sim 1°$. The spectrum C_ℓ^{TT} shows a series of acoustic peaks whose positions and amplitudes depend on $\Omega_\mathrm{m}, \Omega_b, \Omega_\Lambda, H_0, \ldots$; fitting the theoretical model to the observed spectrum is what determines the parameters.

1.3 The displayed fields, one by one

1.3.1 The z field and the presets

In the program

The z field accepts values between 0 and 1500, with five decimals; both "." and "," work as decimal separator. The **preset** buttons load:

- **M 87** ($z = 0.00428$): the giant elliptical galaxy of the Virgo cluster, famous for hosting the first direct image of a supermassive black hole (EHT, 2019).
- **3C 273** ($z = 0.158$): the brightest quasar known, and one of the nearest.
- **z = 1**: a teaching landmark, the typical limit of deep optical surveys.
- **z = 2.34**: the running example used throughout this course.
- **ULAS J1120** ($z = 7.085$): one of the most distant quasars known (Mortlock et al. 2011).
- **GN-z11** ($z = 10.6$): a galaxy observed by JWST in 2022, when the universe was only ~ 435 Myr old.
- **Reionisation** ($z = 20$): the epoch when the first stars reionise the neutral universe.
- **CMB** ($z = 1089.80$): the last-scattering surface.

1.3.2 Looking up an object by name

In the program

The **Object** field saves having to remember redshifts: a name is typed freely (`m31`, `ngc224`, `3c273`, `Sombrero`, `GN-z11`) and the question goes to **SIMBAD**, the astronomical database of the Strasbourg astronomical Data Center. Case, spaces and hyphens do not matter.

Two cautions are worth keeping in mind, as they belong to the physics as much as to the software.

A redshift is not always a distance. Below $z \approx 0.03$, an object's own motion inside its cluster — a few hundred $\mathrm{km\,s^{-1}}$ — remains comparable to the expansion velocity, so a "distance" derived from redshift is merely indicative there. M 31 is the extreme case, at $z = -0.001$: the Andromeda galaxy is approaching us at $300\,\mathrm{km\,s^{-1}}$, and its cosmological distance simply has no meaning. The program says so instead of computing.

Homonyms exist. `GN-z11` is the distant galaxy at $z = 10.6$; `GNz11`, without the hyphen, is object no. 11 of the GNZ catalogue, a nearby galaxy at $z = 0.053$. SIMBAD answers exactly what it is asked: the identifier that replied is shown next to the field, and it is the one to read.

1.3.3 Comoving distance

In the program

Quick definition. The "present-day" distance between us and the object, measured in the frame that expands with the universe. It is the distance *as it is today*, after expansion has carried the object away from us.

Rigorous definition and derivation: chapter 7.

1.3.4 Luminosity distance

In the program

Quick definition. The distance one must put into the classical flux formula $F = L/(4\pi D^2)$ to relate the observed flux F to the intrinsic luminosity L of the source, despite the expansion. It is larger than the "real" distance because expansion makes sources look fainter than they would in a static universe.

Rigorous definition and derivation: chapter 8.

1.3.5 Angular diameter distance

In the program

Quick definition. The distance one must put into $\theta = \ell/D$ (angular size = physical size / distance) to relate the observed angular size to the physical size of the object.

The surprise. At high z ($z > 1.6$) this distance *decreases* with z! A more distant object can look *larger* than a nearer one.

Rigorous definition and derivation: chapter 9.

1.3.6 Light-travel distance

In the program

Quick definition. The *intuitive* value: "the light took such and such a time to arrive, so the object is at c × that time light-years". Bounded by $c\,t_0 \approx 13.79$ Gly (age × c).

But. This distance has no proper metric meaning: it is neither the present-day (comoving) distance nor the distance at emission (angular diameter). It is merely the travel time times the speed of light. We give it because it is so commonly quoted.

Rigorous definition: chapter 10.

1.3.7 Lookback time

In the program

The *lookback time* is the time (in years) elapsed since the light we receive today left the source. For the CMB it is ~ 13.79 Gyr; for a galaxy at $z = 1$, ~ 7.9 Gyr.

Relation to the light-travel distance: $D_{\mathrm{lt}} = c \times t_L$.

Rigorous definition: equation (10.1).

1.3.8 Age of the universe at z

In the program

The age the universe had *when the light was emitted*, counted from the Big Bang ($t = 0$). For the CMB it is $\sim 372\,000$ years; for $z = 2.34$, ~ 2.8 Gyr.

The identity "lookback + age at z = present age ($t_0 = 13.79$ Gyr)" is therefore satisfied on every line.

1.3.9 Scale factor $a = 1/(1+z)$

In the program

Recall: $a(t)$ is the dimensionless factor measuring the *relative size* of the universe at time t, normalised so that $a(t_0) = 1$ today.

For light emitted at redshift z, $a_{\rm em} = 1/(1+z)$.

So at $z = 2.34$: $a_{\rm em} = 0.2994$. The universe was about 3.34 times smaller (in every direction) when the light was emitted.

1.3.10 $E(z) = H(z)/H_0$

In the program

The expansion rate at the epoch of emission, relative to today. This is the function that gets integrated for *all* the distances; the program displays it so that one can see where the numbers come from. At $z = 2.34$: $E = 3.5056$, i.e. $H(z) = 237.2$ km/s/Mpc. At z_*: $E = 23\,053$. Its exact form is given in (6.1).

1.3.11 The error bars

In the program

Every distance and every duration is displayed as *value* ± *uncertainty*, propagated from $\sigma(H_0) = 0.42$ km/s/Mpc and $\sigma(\Omega_{\rm m}) = 0.0056$ by numerical partial derivatives, **taking the correlation between the two into account** (section 6.5):

$$\sigma_G^2 = A^2\sigma_{H_0}^2 + B^2\sigma_{\Omega_{\rm m}}^2 + 2\rho\, A\, B\, \sigma_{H_0}\sigma_{\Omega_{\rm m}}, \qquad A = \frac{\partial G}{\partial H_0}, \quad B = \frac{\partial G}{\partial \Omega_{\rm m}}.$$

At $z = 2.34$: $D_C = 18.824 \pm 0.033$ Gly, i.e. ± 0.17 %. The status bar also shows what the same calculation would give *without* the correlation (± 0.79 %): the factor of 4.5 between the two is not a presentational detail.

1.3.12 Curvature Ω_k

In the program

The **Curvature** Ω_k field (range ± 0.05, default 0) allows the flat universe to be left behind. As soon as $\Omega_k \neq 0$ a fifth row appears: the **transverse comoving distance** D_M, given by (7.3) — sinh if $\Omega_k > 0$, sin if $\Omega_k < 0$. It is that one, not D_C, which then enters $D_L = (1+z)D_M$ and $D_A = D_M/(1+z)$.

The "flat ($\Omega_k = 0$)" button restores Planck 2018. Note that the present age t_0 also changes with curvature (13.744 Gyr for $\Omega_k = +0.01$), and that the status-bar check "$t_L + t_{\rm em} = t_0$" refers to the age of the *current* model.

1.3.13 The SH0ES comparison

In the program

The **"compare with SH0ES"** checkbox adds, under each value, the same quantity computed with $H_0 = 73.04$ km/s/Mpc (the local measurement, Riess et al. 2022) together with the relative difference; the plot overlays the corresponding dotted curves.

Every distance shrinks by 7.4 %, and the durations with them. Compare that with the 0.2 % error bars: **the dominant uncertainty on a cosmological distance today is not statistical, it is systematic.** That is precisely what the Hubble tension is about.

1.3.14 Recession velocities — all three

In the program

The program displays **three** values, because they differ radically and two of them are approximations:

Definition	**at** $z = 2.34$	**status**
naive Doppler $v = cz$	701 514 km/s = 2.340 c	valid if $z \ll 1$
relativistic Doppler	250 467 km/s = 0.835 c	inappropriate in cosmology
FLRW $v = H_0 D_C$	390 497 km/s = 1.303 c	**the right one**

The third one also exceeds c, and that is perfectly legitimate: it is not a propagation speed *through* space but the stretching rate of space itself. See chapter 16.

1.3.15 The status-bar checks

In the program

The status bar permanently displays two consistency checks: $t_L(z) + t_{\mathrm{em}}(z) = t_0$ (satisfied to within 10^{-4} μGyr) and the Etherington identity $D_L/[(1+z)^2 D_A] = 1$ (satisfied to machine precision). If either drifts, the backend has changed version or cosmology.

1.4 The log-log plot

The horizontal axis is the redshift z on a logarithmic scale; the vertical axis is distance in Gly, also logarithmic. The four curves (comoving, luminosity, angular diameter, light travel) are precomputed on $z \in [10^{-3}, 1500]$ and cached on disk.

Adaptive zoom: the visible window follows the current z as $[z/30,\ 8z]$, which keeps both nearby galaxies and the CMB legible.

The dotted vertical lines mark $z = 1$, the maximum of D_A (which follows Ω_k), GN-z11 ($z = 10.6$) and the CMB ($z = 1089.8$); the dash-dotted horizontal lines mark $c\,t_0$ and the particle horizon.

Chapter 2

The preselected targets

This chapter describes each of the eight clickable *presets*, their scientific interest and their place in the history of observational cosmology.

2.1 M 87 — the supermassive galaxy in Virgo ($z = 0.00428$)

M 87 (NGC 4486) is a giant elliptical galaxy at the centre of the Virgo cluster, about 55 Mly away. Its redshift is very small because it belongs to our immediate cosmic neighbourhood. It hosts a supermassive black hole (M 87*) of about $6.5 \times 10^9\ M_\odot$, of which the *Event Horizon Telescope* published the first direct image in 2019 [22]. It also emits a relativistic extragalactic jet observed over several kpc.

Why as a preset? To anchor the user in the *local* universe: at such a small z the four cosmological distances become nearly indistinguishable and all equal $cz/H_0 = 18.96$ Mpc $= 61.8$ Mly (the exact values range from 61.53 Mly for D_A to 62.06 Mly for D_L).

Warning — conceptual trap

Do not confuse Mpc and Mly. $cz/H_0 = 18.96$ Mpc, that is 61.8 Mly and not "19 Mly".

And above all: that distance is wrong for M 87. Direct measurements (surface-brightness fluctuations, Cepheids) give ~ 55 Mly. The ~ 12 % difference is not a computational error: within the Virgo cluster the *peculiar velocity* of galaxies (several hundred km/s, caused by the cluster's own gravity) adds to the Hubble flow and pollutes the redshift. Converting z into a distance only makes statistical sense beyond $z \sim 0.03$, where the Hubble flow dominates peculiar motions. This is the fundamental limitation of this program in the local universe.

2.2 3C 273 — the first quasar ($z = 0.158$)

3C 273 is the 273$^{\text{rd}}$ object of the *Third Cambridge Catalogue of Radio Sources*. In 1963 Maarten Schmidt, at the Mount Palomar telescope, recognised its emission lines as hydrogen Balmer lines shifted by a factor 1.158. At visual magnitude ~ 12.9 it is the brightest quasar seen from Earth [23]. Its bolometric luminosity reaches $\sim 4 \times 10^{46}$ erg/s, i.e. $\sim 10^{13}$ solar luminosities. At $z = 0.158$ the effects of expansion start to show: $D_C = 2.197$ Gly, $D_L = 2.544$ Gly and $D_A = 1.897$ Gly — three numbers already differing by 34 % between the extremes, while z is still small.

2.3 $z = 1$ — a cosmological landmark

At $z = 1$ the universe was about 5.851 Gyr old (a little less than half its present age). This is the typical limit of deep optical galaxy surveys (before JWST). The scale factor was $a = 0.5$, so the universe was *twice smaller* in every direction.

2.4 $z = 2.34$ — "cosmic noon"

"Cosmic noon" denotes the epoch when star formation and quasar activity peaked in the universe (typically $z \approx 1.5$ to 3). The BOSS/eBOSS spectroscopic surveys target that slice preferentially, and $z = 2.34$ is a reference point used in baryon acoustic oscillation (BAO) analyses through quasars and the Lyman-α forest [26].

2.5 ULAS J1120+0641 — a quasar 749 Myr after the Big Bang ($z = 7.085$)

Discovered in 2011 by the UKIDSS infrared survey, ULAS J1120+0641 was for several years the most distant known quasar [24]. Its central black hole weighs $\sim 2 \times 10^9\ M_\odot$ although the universe was only 749 Myr old (the paper quotes 770 Myr, obtained with the WMAP7 cosmology of the time; the 3 % difference illustrates nicely that "the age of an object" is a *model-dependent* quantity). How such a massive black hole formed so quickly remains one of the great theoretical challenges (direct collapse? super-Eddington accretion? primordial seeds?).

2.6 GN-z11 — the galaxy at 435 Myr ($z = 10.6$)

GN-z11 was first identified photometrically by Hubble in 2016 (Oesch et al.), then confirmed spectroscopically by JWST in 2022/2023 [25]. At $z = 10.603$ the universe was only ~ 435 Myr old and GN-z11 was already forming stars at a high rate. It remains one of the most distant confirmed objects (though JWST keeps finding more distant ones).

2.7 Reionisation — $z \approx 6$ to 20

This preset is not an object but an *epoch*. After recombination (at $z \approx 1100$) the hydrogen of the universe is neutral and opaque to stellar UV photons. When the first stars ignite (Pop III, $z \sim 30$), their UV radiation progressively reionises the intergalactic hydrogen. The end of this *epoch of reionisation* (EoR) lies around $z \approx 5.5$-6. It is detected indirectly through:

- the CMB optical depth ($\tau \approx 0.054$, Planck 2018);
- Lyman-α forest absorption in background quasars;
- the 21 cm line of neutral hydrogen (HERA, and SKA in progress).

2.8 CMB — the last-scattering surface ($z_* = 1089.80$)

The *Cosmic Microwave Background* is the oldest radiation we can detect directly. It comes from the epoch when the cooling universe became transparent for the first time: free electrons and protons recombined into neutral hydrogen, releasing the photons that have travelled ever since. See the dedicated chapter 12.

Part II

Theoretical framework

Chapter 3

The cosmological principle and the FLRW metric

3.1 The cosmological principle

Level 1 — Plain language

The idea. On very large scales (beyond a third of a billion light-years), the universe looks the same everywhere and in every direction. No centre, no edge, no privileged region. Clusters of galaxies look statistically alike wherever one looks. This is an observation, not a gratuitous assumption.

Level 2 — Undergraduate

Restatement. We postulate that there exists a frame (that of the *comoving observers*, attached on average to the clusters of galaxies) in which two statistical properties hold on large scales:

- **Homogeneity**: the mean density ρ does not depend on position $\vec{x}$.
- **Isotropy**: ρ, and any observable, does not depend on direction either.

If *every observer* in the universe sees an isotropic universe, then by Schur's theorem the universe is necessarily homogeneous, and the spatial metric must have constant curvature.

Definition 3.1 (Cosmological principle)**.** *There exists a preferred foliation of spacetime into spatial hypersurfaces Σ_t orthogonal to a reference congruence (the* fundamental*, or* comoving*, observers), such that the induced metric on each Σ_t is **homogeneous** and **isotropic**.*

3.2 The Friedmann–Lemaitre–Robertson–Walker metric

Level 1 — Plain language

The rubber-sheet picture. Consider a large rubber sheet with galaxies drawn on it as dots. Stretching the sheet uniformly in every direction leaves the dots fixed *on the sheet* while they move apart from one another. The "stretch factor" as a function of time is called $a(t)$. All of cosmic expansion is contained in that simple idea.

Level 2 — Undergraduate

Spatially flat FLRW metric. In *comoving* Cartesian coordinates (x, y, z) (which do not change in time for a fundamental observer), the spatial line element at time t reads

$$\mathrm{d}s^2_{\text{spatial}} = a^2(t)\,(\mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2),$$

i.e. Pythagoras with a factor $a(t)^2$ that dilates all distances with time. Adding the time dimension:

$$\mathrm{d}s^2 = -c^2\,\mathrm{d}t^2 + a^2(t)\,(\mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2).$$

The minus sign on the time term reflects that we are doing relativity (Lorentzian spacetime). In comoving spherical coordinates (χ, θ, ϕ):

$$\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + a^2(t)\,(\mathrm{d}\chi^2 + \chi^2\,(\mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\phi^2)).$$

A maximally symmetric (homogeneous and isotropic) Riemannian 3-manifold admits exactly three spatial geometries: Euclidean space $\mathbb{E}^3$, the sphere S^3, or hyperbolic space $\mathbb{H}^3$. Their common metric reads, in generalised spherical coordinates (χ, θ, ϕ) centred on the observer:

$$\mathrm{d}\sigma^2 = \mathrm{d}\chi^2 + S_k^2(\chi)\,\mathrm{d}\Omega^2, \quad \mathrm{d}\Omega^2 = \mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\phi^2, \tag{3.1}$$

where χ is a dimensionless *comoving radial coordinate* and $S_k(\chi)$ encodes the spatial curvature of sign $k \in \{-1, 0, +1\}$:

$$S_k(\chi) = \begin{cases} \sin\chi & \text{if } k = +1 \text{ (sphere, closed universe)}, \\ \chi & \text{if } k = 0 \text{ (flat space)}, \\ \sinh\chi & \text{if } k = -1 \text{ (open universe)}. \end{cases} \tag{3.2}$$

Multiplying the hypersurfaces Σ_t by a scale factor $a(t)$ and using the cosmic time t (the proper time of comoving observers) yields the Friedmann–Lemaitre–Robertson–Walker (FLRW) metric:

$$\boxed{\mathrm{d}s^2 = -c^2\,\mathrm{d}t^2 + a^2(t)\,[\mathrm{d}\chi^2 + S_k^2(\chi)\,\mathrm{d}\Omega^2].} \tag{3.3}$$

Remark 3.2. *The coordinate χ is* comoving*: a galaxy that follows the Hubble flow keeps a constant χ. All the time dependence is in $a(t)$. This is the formal statement of "the galaxies do not move, space expands".*

3.3 Comoving radial coordinate and instantaneous proper distance

Level 2 — Undergraduate

Proper distance. At fixed time t ($\mathrm{d}t = 0$), along a radial direction ($\mathrm{d}\Omega = 0$), the FLRW line element gives $\mathrm{d}s = a(t)\,\mathrm{d}\chi$, hence

$$D_p(t) = a(t)\,\chi. \tag{3.4}$$

This is the distance one would measure with a chain of rulers laid end to end *at one instant* of cosmic time. It is not directly observable — no signal travels instantaneously — but it is the cleanest notion conceptually.

Today, $a(t_0) = 1$, so the proper distance coincides with the comoving distance: $D_p(t_0) = \chi$. That is exactly what the program calls the *comoving distance* D_C.

3.4 Recession velocity and the Hubble flow

Differentiating (3.4) at fixed χ:

$$v_{\rm rec}(t) \equiv \dot{D}_p(t) = \dot{a}(t)\,\chi = \frac{\dot{a}(t)}{a(t)}\,D_p(t) \equiv H(t)\,D_p(t), \tag{3.5}$$

which defines the **Hubble parameter** $H(t) \equiv \dot{a}/a$, whose present value is H_0. Equation (3.5) is Hubble's law, exact at all times and all distances within FLRW — with the important caveat that $v_{\rm rec}$ is a rate of change of a proper distance, not a velocity through space (chapter 16).

Chapter 4

Cosmological redshift

4.1 Observational definition

Level 1 — Plain language

What is measured. A galaxy emits light with the spectral lines of the elements it contains — hydrogen, calcium, oxygen. In the observed spectrum, the Hα line is no longer at 656.3 nm: it is shifted towards the red, say to 1969 nm. The redshift is the relative shift: $(1969-656.3)/656.3 = 2.0$. The more distant a galaxy, the larger its redshift.

Definition 4.1 (Redshift)**.**

$$z \equiv \frac{\lambda_{obs} - \lambda_{em}}{\lambda_{em}} = \frac{\lambda_{obs}}{\lambda_{em}} - 1. \tag{4.1}$$

For instance, if the Hα line observed at 1969 nm is emitted at 656.3 nm: $z = (1969 - 656.3)/656.3 = 2.00$. Redshifts are also measured photometrically (from broad-band colours), less precisely.

4.2 Derivation from the FLRW metric

Level 2 — Undergraduate

Light travels on null geodesics. Setting $\mathrm{d}s^2 = 0$ in (3.3) for a radial ray:

$$c\,\mathrm{d}t = -a(t)\,\mathrm{d}\chi \quad \Longrightarrow \quad \chi = \int_{t_{\rm em}}^{t_0} \frac{c\,\mathrm{d}t}{a(t)}. \tag{4.2}$$

Consider two successive wave crests, emitted at $t_{\rm em}$ and $t_{\rm em} + \delta t_{\rm em}$, received at t_0 and $t_0 + \delta t_0$. Both travel the same comoving distance χ, hence

$$\int_{t_{\rm em}}^{t_0} \frac{c\,\mathrm{d}t}{a} = \int_{t_{\rm em}+\delta t_{\rm em}}^{t_0+\delta t_0} \frac{c\,\mathrm{d}t}{a} \quad \Longrightarrow \quad \frac{\delta t_0}{a(t_0)} = \frac{\delta t_{\rm em}}{a(t_{\rm em})}.$$

Since $\lambda = c\,\delta t$ and $a(t_0) = 1$:

$$\boxed{1 + z = \frac{a(t_0)}{a(t_{\rm em})} = \frac{1}{a_{\rm em}}.} \tag{4.3}$$

The redshift is therefore *exactly* the factor by which the universe has expanded since emission. It is not a velocity measurement: it is a measurement of the expansion factor.

Warning — conceptual trap

"The galaxies fly away and their light is Doppler-shifted" is misleading. In FLRW the source is at rest in comoving coordinates. The wavelength stretches *because space stretches while the photon travels*. Locally, the two descriptions agree (a Doppler shift is recovered for $z \ll 1$), but globally the expansion interpretation is the one that generalises. See Bunn & Hogg [18] for a careful discussion.

4.3 Recession velocity and the apparent violation of c

Writing $v = cz$ gives, for $z = 2.34$, $v = 2.34\,c$. This is not a violation of relativity but a misuse of the Doppler formula: see chapter 16, where the three definitions in circulation are compared.

4.4 Other redshifts

A measured redshift is not purely cosmological. It combines:

- the **cosmological** redshift $z_{\rm cosmo}$ from expansion;
- the **Doppler** redshift $z_{\rm D}$ from the peculiar velocity of the source and of the observer (in a cluster, typically ~ 300 km/s, i.e. $z_{\rm D} \sim 10^{-3}$);
- the **gravitational** redshift $z_{\rm grav}$ from the potential well of the source (negligible except near compact objects).

They combine multiplicatively: $(1 + z_{\rm obs}) = (1 + z_{\rm cosmo})(1 + z_{\rm D})(1 + z_{\rm grav})$. This is exactly why M 87's redshift gives a distance 12 % larger than direct measurements.

4.5 Thermal evolution

Since wavelengths scale as a, a blackbody stays a blackbody with $T \propto 1/a \propto (1 + z)$. Hence the CMB temperature today, $T_0 = 2.7255\,\text{K}$, was $T = T_0(1 + z)$ in the past. At $z = 1089$: $T \approx 3000$ K, which is exactly the *recombination* temperature — that is not a coincidence but the definition of z_* (chapter 12).

Chapter 5

The Friedmann equations

5.1 Stress-energy tensor of a perfect fluid

The cosmological principle forces the matter content to be described, on average, by a perfect fluid:

$$T^{\mu\nu} = \left(\rho + \frac{p}{c^2}\right) u^\mu u^\nu + p\, g^{\mu\nu}, \tag{5.1}$$

with ρ the energy density divided by c^2, p the pressure and u^μ the four-velocity of the comoving observers. Each component obeys an equation of state $p = w\rho c^2$:

Component	w	**Dilution**
non-relativistic matter (baryons, CDM)	0	$\rho \propto a^{-3}$
radiation (photons, relativistic neutrinos)	1/3	$\rho \propto a^{-4}$
cosmological constant Λ	-1	$\rho = \text{const}$

Level 1 — Plain language

Why does radiation dilute faster than matter? Because two effects combine: the number of photons per unit volume falls as a^{-3}, like matter, *and* each photon loses energy as a^{-1} through the redshift. Hence a^{-4} in total. That is why radiation dominated the early universe even though it is negligible today.

5.2 Einstein equations with a cosmological constant

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}, \qquad G_{\mu\nu} = R_{\mu\nu} - \tfrac{1}{2} R\, g_{\mu\nu}. \tag{5.2}$$

Injecting the FLRW metric (3.3) into (5.2) gives two independent equations, obtained from the 00 and ij components.

5.3 First Friedmann equation

$$\boxed{\left(\frac{\dot a}{a}\right)^2 = H^2(t) = \frac{8\pi G}{3}\rho - \frac{kc^2}{a^2} + \frac{\Lambda c^2}{3}.} \tag{5.3}$$

Level 2 — Undergraduate

Newtonian analogy. For a test particle on the edge of a homogeneous sphere of radius $R = a\chi$, energy conservation gives $\frac{1}{2}\dot{R}^2 - GM/R = E$, i.e. exactly (5.3) with $-kc^2$ playing the role of the total energy. The analogy is not a derivation (it ignores pressure and relativity) but it makes the equation memorable.

5.4 Second Friedmann equation (acceleration)

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3p}{c^2}\right) + \frac{\Lambda c^2}{3}. \tag{5.4}$$

Note that pressure *slows* the expansion: in general relativity, pressure gravitates. Only a component with $p < -\rho c^2/3$, i.e. $w < -1/3$, can accelerate the expansion — which is why dark energy needs $w \approx -1$.

5.5 Continuity equation

Combining the two Friedmann equations (or using $\nabla_\mu T^{\mu\nu} = 0$):

$$\dot{\rho} + 3\frac{\dot{a}}{a}\left(\rho + \frac{p}{c^2}\right) = 0. \tag{5.5}$$

With $p = w\rho c^2$ and constant w this integrates to $\rho \propto a^{-3(1+w)}$, which reproduces the dilution table above.

5.6 Dimensionless form

Dividing (5.3) by H_0^2 and defining the density parameters $\Omega_i \equiv \rho_i/\rho_{c,0}$ with $\rho_{c,0} = 3H_0^2/8\pi G$:

$$\boxed{\frac{H^2(z)}{H_0^2} = \Omega_\mathrm{r}\,(1+z)^4 + \Omega_\mathrm{m}\,(1+z)^3 + \Omega_k\,(1+z)^2 + \Omega_\Lambda \equiv E^2(z).} \tag{5.6}$$

This single function $E(z)$ is what every distance integral in this course uses. Its exact content in the program is detailed in section 6.3.

Chapter 6

The ΛCDM model and the Planck 2018 parameters

6.1 Definition of the model

Level 1 — Plain language

Λ**CDM** is the "standard model" of cosmology. It says:

1. hot Big Bang, expansion governed by general relativity;
2. **C**old **D**ark **M**atter: massive, slow, collisionless particles that only interact gravitationally;
3. dark energy represented by a cosmological constant Λ that accelerates the expansion;
4. spatially flat universe.

Six parameters suffice. It is the simplest model that fits *all* the observations.

6.2 Planck 2018 values

Parameter	Symbol	Value
Hubble constant today	H_0	$67.66 \pm 0.42\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$
Reduced Hubble parameter	h	0.6766
Total matter density	Ω_m	0.3111 ± 0.0056
Baryon density	$\Omega_b h^2$	0.02242 ± 0.00014
CDM density	$\Omega_c h^2$	0.11933 ± 0.00091
Dark energy density	Ω_Λ	0.6889 ± 0.0056
Curvature density	Ω_k	≈ 0 (within $\sim 2\sigma$)
Photon density	Ω_γ	5.4020×10^{-5} (from T_0)
Equivalent radiation density	Ω_r	9.139×10^{-5} $= \Omega_\gamma(1 + \frac{7}{8}(\frac{4}{11})^{4/3} N_\mathrm{eff})$
Neutrinos today	Ω_ν	1.4397×10^{-3} (non-relativistic)
Effective species	N_eff	3.046
CMB temperature	T_0	2.7255 ± 0.0006 K
Age of the universe	t_0	13.787 ± 0.020 Gyr
CMB redshift	z_*	1089.80 ± 0.21

6.3 The $E(z)$ actually computed — and why $\Omega_\mathrm{m} = 0.3111$ is not astropy's Ω_m

This is the point most often glossed over, including by online calculators. The program does *not* use the two-term formula of popular accounts, but:

$$\boxed{E^2(z) = \Omega_\mathrm{r}(z)\,(1+z)^4 + \Omega_\mathrm{m}^{\mathrm{astropy}}(1+z)^3 + \Omega_k(1+z)^2 + \Omega_\Lambda} \tag{6.1}$$

with two subtleties:

1. **The Planck paper's Ω_m includes the neutrinos; astropy's does not.** `Planck18` carries $\Omega_\mathrm{m}^{\mathrm{astropy}} = 0.30966$ (baryons + dark matter only) and $\Omega_\nu = 0.0014397$ separately. Their sum is 0.31110: exactly the "$\Omega_\mathrm{m} = 0.3111$" of Planck's Table 2. Neutrinos count as *matter* there because, with $\sum m_\nu = 0.06\,\mathrm{eV}$, they are non-relativistic today.

2. Ω_r **depends on** z, because neutrinos change nature: non-relativistic at $z = 0$ (diluting as $(1+z)^3$), relativistic beyond $z \sim 200$ (diluting as $(1+z)^4$). astropy writes $\Omega_\mathrm{r}(z) = \Omega_\gamma[1 + \rho_\nu(z)/\rho_\gamma(z)]$ with

$$\frac{\rho_\nu}{\rho_\gamma} = \frac{7}{8}\left(\frac{4}{11}\right)^{4/3}\frac{N_\mathrm{eff}}{3}\sum_i f\left(\frac{m_i c^2}{k_B T_\nu(z)}\right), \qquad f(y) = \frac{120}{7\pi^4}\int_0^\infty \frac{x^2\sqrt{x^2+y^2}}{e^x+1}\,\mathrm{d}x,$$

f being evaluated through the analytic fit of Komatsu et al. (2011).

Worked example

What does the simplified formula cost? Comparing $E_\mathrm{simple} = \sqrt{\Omega_\mathrm{m}(1+z)^3 + \Omega_\Lambda}$ (with $\Omega_\mathrm{m} = 0.3111$) to the full E:

z	full E	simplified E	difference
1	1.78289	1.78261	−0.016 %
2.34	3.50560	3.50434	−0.036 %
10.6	22.0837	22.0518	−0.14 %
100	573.82	566.15	−1.34 %
1089.8	23 053	20 094	**−12.8** %

Consequences for the observables: $D_C(z_*)$ goes from 45.28 to 45.49 Gly (+0.46 %, harmless), but the **age of the universe at the CMB goes from** 372 **kyr to** 479 **kyr**, i.e. +29 %. In other words: for distances the two-term formula is fine; for *ages* at high z it is plainly wrong, because the age integral is dominated by the radiation era.

6.4 Observed curvature

Planck 2018 measures $\Omega_k = 0.0007 \pm 0.0019$ (consistent with zero at $\sim 0.4\sigma$). Combining CMB+BAO gives $|\Omega_k| < 0.001$. All the calculations in this course adopt $\Omega_k = 0$, but the program allows $\Omega_k \neq 0$ to be explored (section 7.3).

6.5 Uncertainties and parameter correlation

Level 1 — Plain language

The idea. The parameters of the model are not known exactly, so neither are the distances. But one cannot simply add up the errors: the parameters are *tied together* by the measurement. An error on H_0 comes with a compensating error on Ω_m, and the distance moves far less than one would think.

6.5.1 Two-parameter propagation

Level 2 — Undergraduate

For a quantity $G(H_0, \Omega_\mathrm{m})$:

$$\sigma_G^2 = A^2\sigma_{H_0}^2 + B^2\sigma_{\Omega_\mathrm{m}}^2 + 2\rho\, A\, B\, \sigma_{H_0}\sigma_{\Omega_\mathrm{m}}, \qquad A = \frac{\partial G}{\partial H_0}, \quad B = \frac{\partial G}{\partial \Omega_\mathrm{m}}, \tag{6.2}$$

where ρ is the correlation coefficient between H_0 and Ω_m. The third term vanishes *only* if the two measurements are independent — which they are not.

6.5.2 Where $\rho = -0.976$ comes from

One does not need Planck's full covariance matrix: it is enough to notice that the combination

$$\omega_m \equiv \Omega_\mathrm{m} h^2 = 0.14239 \pm 0.00092$$

is far better measured (0.65 %) than Ω_m (1.8 %) or h (0.62 %) separately. This is the signature of the *geometric degeneracy*: the CMB primarily constrains ω_m (which sets the acoustic scale), and separating Ω_m from h comes only afterwards. Propagating (6.2) on ω_m:

$$\left(\frac{\sigma_\omega}{\omega}\right)^2 = \left(\frac{\sigma_{\Omega_\mathrm{m}}}{\Omega_\mathrm{m}}\right)^2 + 4\left(\frac{\sigma_h}{h}\right)^2 + 4\rho\,\frac{\sigma_{\Omega_\mathrm{m}}}{\Omega_\mathrm{m}}\frac{\sigma_h}{h} \implies \boxed{\rho = -0.9763}$$

(symbolic solution and numerical application in `verif_sage/verif_courbure_sigma.sage`).

6.5.3 What difference it makes

	$\sigma(D_C)/D_C$		$\sigma(t_\mathrm{em})/t_\mathrm{em}$	
z	with ρ	without ρ	with ρ	without ρ
0.01000	0.62 %	0.62 %	0.16 %	0.80 %
0.10	0.58 %	0.62 %	0.14 %	0.83 %
0.50	0.44 %	0.65 %	0.17 %	0.94 %
1.00	0.31 %	0.70 %	0.24 %	1.01 %
2.34	0.17 %	0.79 %	0.30 %	1.07 %
5.00	0.13 %	0.86 %	0.32 %	1.09 %
10.60	0.14 %	0.90 %	0.32 %	1.08 %
100.0	0.17 %	0.94 %	0.30 %	1.03 %
1089.8	0.18 %	0.95 %	0.21 %	0.69 %

Two lessons:

1. **Ignoring ρ overestimates the uncertainty by a factor of 2 to 5.** On D_C at $z = 2.34$: ± 0.17 % instead of ± 0.79 %.

2. **There is a "pivot" redshift.** The relative uncertainty on D_C decreases from $z = 0$ (± 0.62 %, governed by H_0 alone, since $D \simeq cz/H_0$) down to $z \approx 5$ (± 0.13 %), then rises again slightly. Around that pivot the model predicts distances better than it knows its own parameters.

Warning — conceptual trap

These error bars do not contain everything. They ignore $\sigma(\Omega_k)$, $\sigma(T_0)$, $\sigma(N_{\text{eff}})$ and $\sigma(\sum m_\nu)$ — negligible here — but above all they say nothing about **model error**: if ΛCDM is wrong, no ± will reveal it. And they are dwarfed by the Hubble tension (next section), which shifts everything by 7 %.

6.6 The Hubble tension

Warning — conceptual trap

A $\sim 5\sigma$ tension today. Local measurements (the distance ladder based on Cepheids and type Ia supernovae, the SH0ES programme [9]) give

$$H_0^{\text{loc}} = 73.04 \pm 1.04 \text{ km s}^{-1}\,\text{Mpc}^{-1},$$

about 5σ above the Planck value 67.66. This tension is one of the major open questions of contemporary cosmology. No new physics is currently accepted as the explanation.

In the program, the "compare with SH0ES" checkbox shows the consequence: every distance shrinks by 7.4 %, every age with it.

Part III

Cosmological distances

Chapter 7

Comoving distance

7.1 Hubble distance

Level 2 — Undergraduate

We define a "natural unit" of cosmological distance:

$$D_H \equiv \frac{c}{H_0}.$$

Numerically, with $H_0 = 67.66$ km/s/Mpc, $D_H = 4430.87$ Mpc $= 1.3587$ Gpc $= 14.4516$ Gly. It is exactly the distance at which the FLRW recession velocity $v = H_0 D$ reaches c: the *Hubble sphere*.

$$D_H \equiv \frac{c}{H_0} = 4430.87 \text{ Mpc} = 1.3587 \text{ Gpc} = 14.4516 \text{ Gly}. \tag{7.1}$$

Warning — conceptual trap

Unit trap. 1 Mpc $= 3.2616 \times 10^6$ ly, so 4430.87 Mpc is *not* 4.43 Gly but 14.45 Gly. Confusing the two divides every distance by 3.26; it is the most common mistake in hand calculations. The ratio D_C/z tends to $D_H = 14.45$ Gly as $z \to 0$, not to 4.4.

7.2 Radial comoving distance

Level 1 — Plain language

The idea. We want the "present-day" distance separating us from the distant object, as if we could freeze the universe now and stretch an infinitely long ruler. That is the comoving distance.

Level 2 — Undergraduate

Key formula.

$$D_C(z) = D_H \int_0^z \frac{dz'}{E(z')} \underset{\substack{\text{neglecting}\\ \text{radiation}}}{\simeq} \frac{c}{H_0} \int_0^z \frac{dz'}{\sqrt{\Omega_m(1+z')^3 + \Omega_\Lambda}}.$$

The integral has no closed form but is trivial numerically (a few milliseconds by Simpson's rule).

Careful, the second expression is an *approximation*: the full $E(z)$ also contains radiation, see (6.1) and section 6.3. It is excellent up to $z \sim 10$ (error < 0.04 % on D_C) but off by 0.46 % at z_* — and above all it would give an age of the universe at the CMB of 479 kyr instead of 372 kyr.

Definition 7.1. *The* radial comoving distance D_C *of a source at redshift z is the* present-day *$(t = t_0)$ distance separating it from the observer, measured along the line of sight at fixed cosmic time.*

From (4.2) and (4.3) (changing variable from t to z through $\mathrm{d}t = -\mathrm{d}a/(aH) = \mathrm{d}z/[(1+z)H(z)]$):

$$\boxed{D_C(z) = D_H \int_0^z \frac{\mathrm{d}z'}{E(z')}} \tag{7.2}$$

7.3 Transverse comoving distance

$$D_M(z) = \begin{cases} \dfrac{D_H}{\sqrt{\Omega_k}} \sinh\left[\sqrt{\Omega_k}\, D_C(z)/D_H\right] & \Omega_k > 0, \\ D_C(z) & \Omega_k = 0, \\ \dfrac{D_H}{\sqrt{|\Omega_k|}} \sin\left[\sqrt{|\Omega_k|}\, D_C(z)/D_H\right] & \Omega_k < 0. \end{cases} \tag{7.3}$$

In Planck 2018, $\Omega_k \approx 0$, hence $D_M = D_C$.

In the program

The program implements all three cases. As long as the "Curvature Ω_k" field is zero, D_M is not displayed since it coincides with D_C; as soon as one moves away from zero, it appears. Some values at $z = 2.34$ (verified in SageMath to five decimals):

Ω_k	D_C (Gly)	D_M (Gly)	D_M/D_C
-0.01	18.8928	18.8390	0.99715
0	18.8240	18.8240	1
$+0.01$	18.7560	18.8087	1.00281

Note the partial cancellation: at $\Omega_k = +0.01$, D_C *decreases* by 0.36 % (the dark-energy content changes) while the hyperbolic geometry *increases* D_M by 0.28 % relative to D_C. In total D_M moves by only 0.08 %: that is why curvature is so hard to measure from distances alone.

Chapter 8

Luminosity distance

8.1 Definition and motivation

Level 1 — Plain language

The practical problem. In astronomy we know the "true" luminosity of certain objects (standard candles such as type Ia supernovae). We measure the flux that reaches us. We would like to deduce the distance. The classical formula $F = L/(4\pi D^2)$ would work... in a static universe. In an expanding one we need an *effective* distance that keeps the formula valid. That is the luminosity distance.

Level 2 — Undergraduate

Three effects dilute the flux.

1. *Geometric spreading.* The radiation spreads over a sphere of area $4\pi a_0^2 D_M^2$.
2. *Energy loss per photon.* $E_\gamma \to E_\gamma/(1+z)$.
3. *Time dilation.* The photon arrival rate is divided by $(1+z)$.

In total $F = L/[4\pi D_M^2\,(1+z)^2]$, hence $D_L = (1+z)\,D_M$.

Definition 8.1.

$$D_L \equiv \sqrt{\frac{L}{4\pi F}}. \tag{8.1}$$

8.2 Derivation from the FLRW metric

(i) Geometric spreading. At time t_0, the photons from a source at comoving distance D_C are spread over a sphere of proper area $4\pi a_0^2 D_M^2(z)$.

(ii) Energy shift. Each photon loses a factor $(1+z)$ of energy.

(iii) Time dilation. The arrival rate of photons is slowed by a further factor $(1+z)$.
In total:

$$F = \frac{L}{4\pi D_M^2(z)\,(1+z)^2}. \tag{8.2}$$

$$\boxed{D_L(z) = (1+z)\,D_M(z).} \tag{8.3}$$

8.3 Distance modulus and supernovae

Level 1 — Plain language

How the acceleration of the expansion was discovered. In 1998-1999 two teams observed type Ia supernovae (which nearly all share the same absolute luminosity) at $z \sim 0.5$. They looked *fainter* than expected in a decelerating universe, hence *more distant*. The only explanation: the expansion accelerates. Nobel Prize 2011 to Riess, Schmidt and Perlmutter.

$$\mu(z) \equiv m - M = 5\log_{10}(D_L/\mathrm{Mpc}) + 25. \tag{8.4}$$

Chapter 9

Angular diameter distance

9.1 Definition

Level 1 — Plain language

The idea. If I know the *physical* size of an object (in km) and I see its *angular* size (in arcsec), the distance linking the two is the angular diameter distance: $\theta = \ell/D_A$.

Level 2 — Undergraduate

Key formula. Since the light was emitted when the universe was smaller by a factor $(1+z)$:

$$D_A(z) = \frac{D_M(z)}{1+z} = \frac{D_C(z)}{1+z} \text{ (in a flat universe).}$$

Definition 9.1.

$$D_A \equiv \frac{\ell}{\theta}. \tag{9.1}$$

9.2 Derivation

$$\boxed{D_A(z) = \frac{D_M(z)}{1+z}.} \tag{9.2}$$

9.3 The maximum of D_A

Warning — conceptual trap

Counter-intuitive! In ΛCDM, $D_A(z)$ rises to a maximum near $z \approx 1.59$ and then *decreases*. An object at $z = 5$ will generally have a *larger* angular size than an identical object at $z = 1$. The reason: the object emitted its light at a time when it was physically *close* to us, so it subtends a large angle.

9.4 Etherington's reciprocity theorem

Theorem 9.2 (Etherington, 1933 [8])**.** *For any null geodesic in an arbitrary Lorentzian geometry:*

$$\boxed{D_L(z) = (1+z)^2\, D_A(z).} \tag{9.3}$$

Chapter 10

Lookback time and light-travel distance

10.1 Lookback time

Level 1 — Plain language

The question. How long has the light been travelling to reach us?

Level 2 — Undergraduate

$$t_L(z) = \int_0^z \frac{\mathrm{d}z'}{(1+z')H(z')}.$$

Again an integral in $\mathrm{d}z/E(z)$, this time weighted by $1/(1+z)$.

$$t_L(z) = t_0 - t_{\rm em}(z) = \int_0^z \frac{\mathrm{d}z'}{(1+z')\,H(z')} = \frac{1}{H_0}\int_0^z \frac{\mathrm{d}z'}{(1+z')\,E(z')}. \tag{10.1}$$

10.2 Light-travel distance

$$D_{\rm lt}(z) \equiv c\,t_L(z). \tag{10.2}$$

Warning — conceptual trap

$D_{\rm lt}$ is the most *intuitive* value but it has no proper metric meaning. It is bounded by $c\,t_0 \approx 13.79$ Gly whatever z: even for the CMB ($z = 1089.8$) it does not exceed that bound. The comoving distance, by contrast, can exceed it (the CMB sits at $D_C \approx 45.3$ Gly).

10.3 Age of the universe at z

$$t_{\rm em}(z) = \int_z^\infty \frac{\mathrm{d}z'}{(1+z')\,H(z')}. \tag{10.3}$$

Chapter 11

Summary

11.1 Summary table

Distance	**Formula**	**Use**
Radial comoving D_C	$D_H \int_0^z \frac{\mathrm{d}z'}{E(z')}$	the "present-day" distance of the source
Transverse comoving D_M	D_C if $\Omega_k = 0$	transverse scales, BAO
Luminosity D_L	$(1+z)\,D_M$	photometry, supernovae
Angular diameter D_A	$D_M/(1+z)$	apparent angular sizes
Light travel D_{lt}	$c\,t_L(z)$	popular accounts, $\leq c\,t_0$

11.2 Etherington identity

$$D_L = (1+z)^2\,D_A, \qquad D_L = (1+z)\,D_M, \qquad D_A = D_M/(1+z).$$

11.3 Asymptotic behaviour

Small z ($z \ll 1$).

$$D_C \simeq \frac{cz}{H_0}\Big[1 - \tfrac{1}{2}(1+q_0)z + \mathcal{O}(z^2)\Big], \qquad D_L \simeq \frac{cz}{H_0}\Big[1 + \tfrac{1}{2}(1-q_0)z + \mathcal{O}(z^2)\Big],$$

with $q_0 = \frac{1}{2}\Omega_{\mathrm{m}} - \Omega_\Lambda = -0.5334$ (Planck 2018).

Large z. D_C tends to the comoving distance to the particle horizon, $\chi_{\mathrm{hor}} = 46.20$ Gly in Planck 2018. Hence $D_A \to \chi_{\mathrm{hor}}/(1+z)$ (decreasing) and $D_L \to \chi_{\mathrm{hor}}(1+z)$ (growing linearly): it is $\chi_{\mathrm{hor}} = 46.2$ Gly that appears, not $D_H = 14.45$ Gly. Check at z_*: $46.20/1090.8 = 0.0424$ Gly, against the exact value $D_A(z_*) = 0.04152$ Gly (the 2 % difference is because $D_C(z_*) = 45.28$ has not quite saturated yet).

Part IV

Observed phenomena: CMB, observable universe, superluminal velocities

Chapter 12

The cosmic microwave background

12.1 What is the CMB?

Level 1 — Plain language

The oldest light. The CMB ("Cosmic Microwave Background") is the oldest light one can see. It fills *all* of space and arrives from every direction at once, in the microwave range (millimetre wavelengths). It was emitted when the universe was $\sim 372\,000$ years old (the Planck 2018 value; the older WMAP figure of 380 000 years is still often quoted), and it has been travelling for ~ 13.8 billion years.

Level 2 — Undergraduate

The CMB is an almost perfect blackbody at $T_0 = 2.7255$ K, with relative fluctuations $\Delta T/T \sim 10^{-5}$ that carry the cosmological information. Its spectrum matches a blackbody to $\sim 10^{-9}$, making it the best measured blackbody spectrum in all of physics.

12.2 Why we still see it today

Level 1 — Plain language

The fog picture. Consider a cloud of warm fog. As long as the cloud is dense, light propagating in it is constantly absorbed and re-emitted: the cloud is *opaque*. When the cloud cools and thins out, at some point the molecules no longer trap the light: the cloud becomes *transparent*. All the trapped light escapes at once, in every direction.

That is what happened to the universe, at $z \approx 1100$. Before that, matter was hot enough to ionise hydrogen into a plasma; photons were trapped by scattering off free electrons (*Thomson scattering*). At about $T = 3000$ K hydrogen became neutral (*recombination*): electrons stuck to protons, and the photons could finally travel freely.

This "release" of photons happens in every region of the universe at the same epoch (to within the fluctuations). So *from all around us*, at 13.79 billion light-years (in light-travel distance), we see this "last-scattering surface". Since the universe has been transparent ever since, that radiation reaches us without further interaction.

Level 2 — Undergraduate

Why $T \approx 3000$ K rather than $T \approx 13.6$ eV $\approx 158\,000$ K (the ionisation energy of hydrogen)? Because photons vastly outnumber baryons (the ratio is $n_\gamma/n_b \approx 1.6 \times 10^9$). Even when the mean energy $k_B T$ has fallen well below 13.6 eV, the Wien tail of the Planck distribution still holds enough UV photons to keep the gas ionised. The Saha calculation puts recombination near $T \approx 3700$ K, refined by non-equilibrium treatment (the Peebles equation) to $T_{\rm rec} \approx 3000$ K, i.e. $z_* = 1089.80$ according to Planck 2018 [1].

12.3 Why exactly $z_* = 1089.80$?

Blackbody radiation cools as $T \propto 1/a \propto (1+z)$. Knowing the present CMB temperature $T_0 = 2.7255\,\mathrm{K}$ and the recombination temperature of hydrogen ($T_{\rm rec} \approx 2970\,\mathrm{K}$ from the Peebles equation):

$$1 + z_* = \frac{T_{\rm rec}}{T_0} = \frac{2970}{2.7255} \approx 1090.$$

So z_* is not an arbitrary number: it is the ratio between the temperature at which the universe became transparent and the present-day temperature of the radiation emitted then.

12.4 Discovery of the CMB and successive missions

1948 Alpher, Herman and Gamow predict a relic radiation at ~ 5 K in the hot Big Bang framework [27].

1965 A. Penzias and R. Wilson accidentally detect an isotropic excess noise at 4080 MHz at Bell Labs (the *Holmdel Horn*). Nobel Prize 1978 [28].

1989-1993 The *COBE* mission (NASA): confirms the perfect blackbody spectrum (FIRAS) and detects the first anisotropies (DMR). Nobel Prize 2006 (Mather, Smoot) [29, 30].

2001-2010 The *WMAP* mission (NASA): high-resolution mapping, constrains H_0, $\Omega_{\rm m}$, Ω_Λ to a few percent.

2009-2013 The *Planck* mission (ESA): the most precise to date, and the source of the parameters used by this program [1].

Chapter 13

Anatomy of the four curves

The program plots the four distances against z simultaneously. At small z they are *all superimposed*, then at large z they diverge spectacularly. This chapter explains why.

13.1 The four equations together

$$\boxed{\begin{aligned}
D_C(z) &= \frac{c}{H_0}\int_0^z \frac{\mathrm{d}z'}{E(z')} && \text{(comoving distance)}\\
D_L(z) &= (1+z)\,D_M(z) && \text{(luminosity distance)}\\
D_A(z) &= \frac{D_M(z)}{1+z} && \text{(angular diameter distance)}\\
D_{\mathrm{lt}}(z) &= c\int_0^z \frac{\mathrm{d}z'}{(1+z')\,H(z')} = \frac{c}{H_0}\int_0^z \frac{\mathrm{d}z'}{(1+z')\,E(z')} && \text{(light travel)}
\end{aligned}} \tag{13.1}$$

where $E(z) = \sqrt{\Omega_{\mathrm{r}}(1+z)^4 + \Omega_{\mathrm{m}}(1+z)^3 + \Omega_k(1+z)^2 + \Omega_\Lambda}$.

13.2 Why they coincide at small z

Level 1 — Plain language

The intuition. At very small z the universe barely changed while the photon travelled. So the precise definition of distance hardly matters: they all tend to the same value, the one given by the usual Hubble law, $D \approx cz/H_0$.

Level 2 — Undergraduate

Leading order. For $z \ll 1$, $E(z) \approx 1$ and $(1+z)^{\pm 1} \approx 1$, so

$$D_C \approx \frac{c}{H_0}\,z, \qquad D_L \approx (1+z)D_C \approx D_C, \qquad D_A \approx D_C/(1+z) \approx D_C,$$

and likewise $D_{\mathrm{lt}} \approx (c/H_0)z \approx D_C$. **All four coincide** at cz/H_0 to first order in z.

13.2.1 Second order (the first departures)

$$D_C(z) = \frac{c}{H_0}\Big[z - \tfrac{1}{2}(1+q_0)\,z^2 + \mathcal{O}(z^3)\Big], \tag{13.2}$$
$$D_L(z) = \frac{c}{H_0}\Big[z + \tfrac{1}{2}(1-q_0)\,z^2 + \mathcal{O}(z^3)\Big], \tag{13.3}$$
$$D_A(z) = \frac{c}{H_0}\Big[z - \tfrac{1}{2}(3+q_0)\,z^2 + \mathcal{O}(z^3)\Big], \tag{13.4}$$
$$D_{\mathrm{lt}}(z) = \frac{c}{H_0}\Big[z - \tfrac{1}{2}(2+q_0)\,z^2 + \mathcal{O}(z^3)\Big], \tag{13.5}$$

where $q_0 = \frac{1}{2}\Omega_{\mathrm{m}} - \Omega_\Lambda$ is the *deceleration parameter* (negative today, since the expansion accelerates):

$$q_0 = \tfrac{1}{2}(0.3111) - 0.6889 = -0.53335 \quad (\text{or } -0.5325 \text{ including radiation, } q_0 = \tfrac{\Omega_{\mathrm{m}}}{2} + \Omega_{\mathrm{r}} - \Omega_\Lambda).$$

Numerically, with $\Omega_{\mathrm{m}} = 0.3111$:

$$D_C/D_H = z - 0.2333\,z^2, \qquad D_L/D_H = z + 0.7667\,z^2,$$
$$D_A/D_H = z - 1.2333\,z^2, \qquad D_{\mathrm{lt}}/D_H = z - 0.7333\,z^2.$$

Warning — conceptual trap

The classic trap. Many texts give $z + \frac{1}{2}(1-q_0)z^2$ for D_C: that is the expansion of the *luminosity* distance (the Weinberg / Hubble–Sandage formula). The four coefficients above follow from one another through the factors $(1+z)^{\pm 1}$ and were re-derived by computer algebra (appendix B); they differ by exactly one unit each, which is visible in the numbers above.

13.3 Why they diverge at large z

Level 2 — Undergraduate

At large z the dominant term of $E(z)$ is $E(z) \sim \sqrt{\Omega_{\mathrm{m}}}\,(1+z)^{3/2}$, so

$$\int_0^z \frac{\mathrm{d}z'}{E(z')} \sim \frac{2}{\sqrt{\Omega_{\mathrm{m}}}}\Big[1 - (1+z)^{-1/2}\Big] \xrightarrow[z\to\infty]{} \frac{2}{\sqrt{\Omega_{\mathrm{m}}}}.$$

D_C **therefore saturates** at a finite value: the comoving distance to the particle horizon, 46.20 Gly.

Caution with that estimate: it assumes matter domination *over the whole range*, which is false at small z (where Λ dominates) and at very large z (where radiation does). It gives $2D_H/\sqrt{\Omega_{\mathrm{m}}} = 51.8$ Gly against the exact 46.20 Gly: good to 12 %, enough to understand *why* the integral converges, not enough to quote a number.

$D_L = (1+z)D_M$ **grows linearly with** z; $D_A = D_M/(1+z)$ **decreases towards zero**; $D_{\mathrm{lt}} \le c\,t_0 \approx 13.79$ Gly **is bounded** (light can travel at most for the age of the universe).

13.3.1 Limiting values for the CMB ($z = 1089.8$)

Distance	Value
$D_C(z_*)$	45.285 Gly
$D_L(z_*)$	49 397 Gly
$D_A(z_*)$	0.04152 Gly = 41.5 Mly
$D_{\mathrm{lt}}(z_*)$	13.7865 Gly

These four numbers, all of which mean "the distance to the CMB", span *six orders of magnitude* (from ~ 40 Mly to $\sim 50\,000$ Gly). That is the divergence one sees on the plot.

13.4 The maximum of D_A near $z = 1.59$

Going further

The angular distance $D_A = D_M/(1+z)$ is the product of an increasing function (D_M) and a decreasing one ($1/(1+z)$). Its derivative vanishes where $\mathrm{d}D_C/\mathrm{d}z = D_C/(1+z)$, i.e. at $z = 1.592133$ in Planck 2018, where $D_A = 5.84629$ Gly. Beyond that, increasing z *reduces* D_A.

Observable consequence. A galaxy of physical size 30 kpc at $z = 5$ appears angularly *larger* than an identical galaxy at $z = 1$. This is counter-intuitive and essential when interpreting JWST images.

Chapter 14

Why redshift is not linear in distance

14.1 The linear Hubble law and its range of validity

Level 1 — Plain language

At small z. For nearby galaxies ($z < 0.1$) we have the **Hubble law**:

$$v \approx H_0 \times D, \qquad cz \approx H_0 \times D.$$

The recession velocity is proportional to distance, and so is the redshift. It is a linear relation, observed by Hubble in 1929. In that regime, doubling the distance doubles the redshift.

But it breaks down far away. That neat linearity rests on two assumptions that fail at large z:

1. that the universe expands at the same rate during the whole light travel. False: $H(z)$ changes with time.
2. that the Doppler shift $v = cz$ applies. False: it is the expansion of space itself that shifts the wavelengths, not motion through space.

14.2 The exact relation

Level 2 — Undergraduate

The true relation between z and comoving distance.

$$D_C(z) = \frac{c}{H_0}\int_0^z \frac{\mathrm{d}z'}{E(z')}, \quad E(z) \simeq \sqrt{\Omega_\mathrm{m}(1+z)^3 + \Omega_\Lambda} \quad \text{(without radiation, cf. (6.1))}.$$

At small z: $E(z) \approx 1$, so $D_C \approx (c/H_0)\,z$, linear.

At large z (matter-dominated regime): $E(z) \sim \sqrt{\Omega_\mathrm{m}}\,(1+z)^{3/2}$, so the integral *saturates*:

$$\int_0^z \frac{\mathrm{d}z'}{(1+z')^{3/2}} = 2\left[1 - (1+z)^{-1/2}\right] \xrightarrow[z\to\infty]{} 2.$$

Conclusion: D_C tends to a *finite* value as $z \to \infty$, called the **particle horizon radius**.

14.3 Numbers for a few redshifts

z	D_C (Gly)	D_C/z (Gly)	cz (km/s)
0.01	0.1442	14.42	2 998
0.1	1.4108	14.11	29 979
0.5	6.3484	12.70	149 896
1	11.0751	11.08	299 792
2.34	18.8240	8.04	701 514
5	25.9173	5.18	1 498 962
10.6	31.8379	3.00	3 177 800
1089.8	45.2848	0.0416	326 713 821
∞	46.2005 (horizon)	0	—

The ratio D_C/z starts at $D_H = 14.45$ Gly at low z and tends to zero: the distance "saturates" while z diverges. The cz column is given for reference only: beyond $z \sim 1$ it exceeds c and is meaningless (see chapter 16).

Chapter 15

The observable universe and its size

15.1 Definition

Level 1 — Plain language

The observable universe is the sphere centred on us containing every region whose light has had time to reach us since the Big Bang. Beyond it, what happens cannot (yet) influence us. So it is a ball, not the whole universe (which may be infinite, or simply much larger).

15.2 Three radii not to be confused

Warning — conceptual trap

One often hears that "the observable universe has a radius of 13.8 billion light-years". **That is wrong.** Three very different numbers must be distinguished.

1. **The distance the light has travelled**: $c\,t_0 = 13.79$ Gly. It is merely the age times the speed of light. **It is not the size of the observable universe**: while the light travelled, space expanded.

2. **The comoving radius of the particle horizon**:

$$D_{\rm hor} = D_H \int_0^\infty \frac{\mathrm{d}z'}{E(z')} = 46.2005 \text{ Gly}.$$

   This is the *present-day* distance to the sphere from which the oldest light comes (in practice the CMB, at $D_C \approx 45.3$ Gly, very close). **This is the "size" of the observable universe in the modern sense.**

3. **The event horizon radius**: 16.58 Gly comoving (computed in appendix B). It is the limit beyond which a signal emitted *today* can never reach us, because of the accelerating expansion. See [20] for a clear discussion.

Level 2 — Undergraduate

Why 46 Gly and not 13.8? While the light travelled, space stretched by a factor $(1+z)$ at each step. The CMB photon was emitted at $D_A \approx 41.5$ Mly (the angular distance, i.e. the physical distance *at emission*) and travelled for 13.79 Gyr, but during that time the emitting matter was carried away by the expansion. Today that same matter sits at $D_C = 45.3$ Gly:

$$D_C = (1+z_*)\,D_A = 1090.8 \times 0.04152 \text{ Gly} \approx 45.28 \text{ Gly}.$$

15.3 Total diameter

The *diameter* of the observable universe is therefore ~ 92.4 Gly, and its *volume* $\sim 4 \times 10^{32}$ cubic light-years. It contains roughly 2×10^{12} galaxies (JWST-era estimate).

15.4 What lies beyond?

Going further

We do not know, by definition. But a few theoretical bounds:

- if the universe is *flat* (Planck 2018: $\Omega_k = 0$ to $\sim 0.5\%$), it may well be *infinite*;
- if one insists on a finite universe, its total size must be at least ~ 250 times the observable part (from the constraint on Ω_k);
- beyond the event horizon, the accelerating expansion driven by Λ will keep new regions *forever* out of reach.

Chapter 16

Recession velocities and the superluminal universe

16.1 The problem

For $z = 2.34$ the program shows a "recession velocity" of $\sim 700\,000$ km/s $= 2.34\,c$ under the naive Doppler definition. For the CMB ($z = 1089.8$) that value would be $\sim 1090\,c$. **What does that mean? Does it violate relativity?**

16.2 Three ways to compute a "recession velocity"

16.2.1 Classical Doppler: $v = cz$

Level 1 — Plain language

This is the everyday Doppler formula: a car receding at 30 m/s shifts a 1000 Hz sound to $\sim$ 999.9 Hz, i.e. $z \approx 10^{-7}$. From that one writes $v = cz$ for light. **It is a very-low-z approximation only.**

For $z = 2.34$: $v_{\text{Doppler}} = 2.34\,c$. That is what the first line of the program shows — as an approximation, not as the "true" velocity.

16.2.2 Relativistic Doppler

In special relativity the shift of a source receding radially at v is

$$1 + z = \sqrt{\frac{1 + v/c}{1 - v/c}}, \qquad \frac{v}{c} = \frac{(1+z)^2 - 1}{(1+z)^2 + 1}.$$

This formula is *bounded by* c (as $z \to \infty$, $v \to c$). For $z = 2.34$: $v/c = (3.34^2 - 1)/(3.34^2 + 1) \approx 0.833$, i.e. 83% of c. For the CMB ($z \to 1090$): $v/c \to 0.999\,998\,3$.

Warning — conceptual trap

The relativistic Doppler formula is conceptually wrong in cosmology. It assumes the source moves *through* a static Minkowski spacetime. But the FLRW universe is not globally flat, and a comoving source does *not* move through space: space stretches between it and us. A different concept is needed.

16.2.3 FLRW recession velocity: $v_{\mathrm{rec}} = H(t)\, D_p(t)$

Level 2 — Undergraduate

The "true" cosmological velocity. For two comoving observers separated by a proper distance $D_p(t) = a(t)\,\chi$ at time t,

$$v_{\mathrm{rec}}(t) = \dot{D}_p(t) = \dot{a}(t)\,\chi = H(t)\,D_p(t).$$

This velocity is *unbounded*: for large enough D_p, $v_{\mathrm{rec}} > c$. But it is not a propagation speed *through* space: it is a *stretching* rate of space itself.

Today, for the CMB:

$$v_{\mathrm{rec}}(t_0) = H_0 \times D_C(\mathrm{CMB}) = 67.66\ \mathrm{km/s/Mpc} \times 13\,884\ \mathrm{Mpc} = 939\,400\ \mathrm{km/s} = 3.134\,c.$$

So even the well-defined recession velocity is superluminal for very distant objects: not only the naive $cz \approx 1090\,c$, but also the cosmologically correct $\sim 3\,c$.

16.3 Why relativity is not violated

Level 1 — Plain language

Einstein's rule, stated strictly. No particle, no signal, no information can propagate *through space* faster than light. Measured by a *local* observer, the speed of an object in its immediate neighbourhood is always $< c$.

But cosmic expansion is not motion through space. It is space itself that dilates. Two comoving observers are *at rest*: their comoving coordinate χ does not change. Only the physical distance between them grows with time. Nothing travels at 1.3×10^6 km/s past anybody.

Level 2 — Undergraduate

Rigorous statement. In general relativity the limit c applies *locally*, within the light cone at each point. Relativity does not forbid two regions of space from separating at an arbitrarily large coordinate velocity. The rule $v \le c$ is a statement of *special* relativity, i.e. of a globally Minkowski spacetime. In cosmology spacetime is curved; velocities between distant points have no unique definition, and the one we use (HD) may exceed c.

16.4 Are superluminal galaxies observable?

Yes. Surprisingly, we do see objects whose recession velocity exceeds c today. The condition for a signal emitted today by an object at D_p to reach us eventually is not $v_{\mathrm{rec}} \le c$ but that D_p be *smaller than the event horizon*. Numerically, $D_{\mathrm{event}} = 16.58$ Gly, whereas the sphere where $v_{\mathrm{rec}}(t_0) = c$ (the Hubble sphere) is at $D_H \approx 14.45$ Gly. So galaxies in the shell between 14.45 and 16.58 Gly recede faster than light yet emit light that will eventually reach us (heavily redshifted). See [20] for details.

Part V

Worked examples

Chapter 17

Numbers for a few emblematic redshifts

17.1 The running example $z = 2.34$

Worked example

Input. $z = 2.34$, Planck 2018 cosmology.

Results.

- $D_C = D_M = 18.824 \pm 0.033$ Gly
- $D_L = (1+z)\,D_M = 62.87 \pm 0.11$ Gly
- $D_A = D_M/(1+z) = 5.6359 \pm 0.0098$ Gly
- $t_L = 10.988$ Gyr, $\quad D_{\rm lt} = c\,t_L = 10.988$ Gly
- $t_{\rm em} = t_0 - t_L = 2.7991$ Gyr
- $a(t_{\rm em}) = 1/(1+z) = 0.2994$
- $E(z) = 3.5056$, i.e. $H(z) = 237.2$ km/s/Mpc
- recession velocity (FLRW) $v = 1.303\,c$

Interpretation. The light was emitted when the universe was 2.80 Gyr old (out of 13.79). At that moment the object was at $D_A = 5.64$ Gly. The light travelled for 10.99 Gyr; meanwhile the universe expanded by a factor 3.34 and the object now lies at $D_C = 18.82$ Gly. D_L reaches 62.87 Gly because of the three cumulative effects.

17.2 The case $z = 1$

Worked example

$D_C = 11.075$ Gly, $D_L = 22.150$ Gly, $D_A = 5.538$ Gly, $t_L = 7.9355$ Gyr, $t_{\rm em} = 5.8513$ Gyr.

17.3 The case $z = 10.6$ (GN-z11)

Worked example

$D_C = 31.838$ Gly, $D_L = 369.32$ Gly, $D_A = 2.7446$ Gly, $t_L = 13.3517$ Gyr, $t_{\text{em}} = 435$ Myr.

GN-z11 is one of the most distant spectroscopically confirmed galaxies (JWST 2022/23); in Planck 2018 cosmology the universe was 435 Myr old. The often quoted "~ 400 Myr" comes from earlier cosmologies and from $z = 11.09$ (the 2016 photometric estimate, corrected to $z = 10.60$ by JWST spectroscopy).

17.4 The case $z = 1089.8$ (CMB)

Worked example

$D_C = 45.285$ Gly, $D_L = 49\,397$ Gly (enormous), $D_A = 0.041\,52$ Gly $= 41.5$ Mly, $t_L = 13.7865$ Gyr, $t_{\text{em}} = 372$ kyr.

The comoving distance to the CMB is very close to the particle horizon radius. The angular distance is only ~ 42 Mly: one causal patch of the CMB subtends $\sim 1^\circ$ on the sky (the first acoustic peak).

Chapter 18

Numerical implementation

18.1 Evaluating the integrals

The integral (7.2) **has no** expression in elementary functions in ΛCDM: the substitution $u = (1+z)^{-3/2}$ leaves an irreducible factor $u^{-2/3}$, and the closed form involves a hypergeometric function (or an incomplete elliptic integral). So it is computed numerically.

What does have a closed form: the age. In flat ΛCDM neglecting radiation ($\Omega_r = 0$), the *age* integral can be done exactly, because the extra weight $1/(1+z)$ makes the substitution complete:

$$\boxed{t(z) = \frac{2}{3\,H_0\sqrt{\Omega_\Lambda}}\ \operatorname{arcsinh}\left[\sqrt{\frac{\Omega_\Lambda}{\Omega_m}}\,(1+z)^{-3/2}\right]} \tag{18.1}$$

Numerical check ($\Omega_m = 0.3111$, $\Omega_\Lambda = 0.6889$, $1/H_0 = 14.4516$ Gyr):

$$t(0) = 13.7915\ \text{Gyr}, \quad t(1) = 5.8555\ \text{Gyr}, \quad t(2.34) = 2.8025\ \text{Gyr},$$

against the exact values (*with* radiation) 13.7869, 5.8513 and 2.7991 Gyr. The agreement is better than 0.04 % as long as $z \lesssim 10$; at z_* the formula gives 479 kyr instead of 372 kyr, because it ignores the radiation era. This closed form was verified symbolically and then numerically (appendix B); it is handy for checking a result of the program by hand.

Warning — conceptual trap

The prefactor is $\frac{2}{3\sqrt{\Omega_\Lambda}}$, not $\frac{2}{\sqrt{\Omega_\Lambda}}$; and this expression gives the *age* $t(z)$, not the comoving distance. The lookback time follows as $t_L(z) = t(0) - t(z)$.

18.2 Python listing

```
from astropy.cosmology import Planck18 as cosmo
from astropy import units as u, constants as const

z = 2.34
D_C  = cosmo.comoving_distance(z).to(u.Glyr)
D_M  = cosmo.comoving_transverse_distance(z).to(u.Glyr)
D_L  = cosmo.luminosity_distance(z).to(u.Glyr)
D_A  = cosmo.angular_diameter_distance(z).to(u.Glyr)
t_L  = cosmo.lookback_time(z).to(u.Gyr)
D_lt = (t_L * const.c).to(u.Glyr)
```

```
age  = cosmo.age(z).to(u.Gyr)

# the three "recession velocities" (see the dedicated chapter)
c_kms  = const.c.to_value(u.km / u.s)
v_cz   = c_kms * z                                   # naive Doppler
v_sr   = c_kms * ((1+z)**2 - 1) / ((1+z)**2 + 1)     # relativistic Doppler
v_flrw = cosmo.H0.value * cosmo.comoving_distance(z).to_value(u.Mpc)   # FLRW
```

Warning — conceptual trap

The giga-light-year unit is spelled `u.Glyr` in `astropy` (from `lyr`), *not* `u.Gly`: the latter does not exist and raises an `AttributeError`. Likewise `u.lyr`, not `u.ly`.

The complete program, with a Qt6 interface reproducing the log-log plot and all four distances as functions of z, is `redshift_distance_gui.py`.

Part VI

Appendices

Appendix A

Physical constants

Constant	Symbol	Value
Speed of light	c	$299\,792.458\,\mathrm{km\,s^{-1}}$ (exact)
Newton's constant	G	$6.674\,30 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$
Parsec	pc	$3.0857 \times 10^{13}\,\mathrm{km}$
Julian year	yr	$3.155\,76 \times 10^{7}\,\mathrm{s}$
Light-year	ly	$9.4607 \times 10^{12}\,\mathrm{km}$
Megaparsec	Mpc	$3.0857 \times 10^{19}\,\mathrm{km} = 3.2616 \times 10^{6}$ ly

Appendix B

Independent verification of the calculations (SageMath)

This course and the program rely on `astropy.cosmology.Planck18`. Trusting a single computation chain is bad practice, so every value has been **recomputed from scratch in SageMath**, without astropy, at extended precision.

B.1 Method

1. **Rebuilding the parameters** from the CODATA 2022 constants and $(H_0, \Omega_\mathrm{m}, T_0, N_\mathrm{eff}, m_\nu)$ alone:
$$\rho_{c,0} = \frac{3H_0^2}{8\pi G}, \quad \Omega_\gamma = \frac{a_B T_0^4}{\rho_{c,0}c^2}, \quad a_B = \frac{\pi^2 k_B^4}{15\hbar^3 c^3}, \quad T_{\nu 0} = \left(\tfrac{4}{11}\right)^{1/3} T_0.$$
2. **Neutrinos through the exact Fermi–Dirac integral** $f(y) = \frac{120}{7\pi^4}\int_0^\infty \frac{x^2\sqrt{x^2+y^2}}{e^x+1}\mathrm{d}x$, instead of the Komatsu et al. (2011) analytic fit used by astropy.
3. **Integrals** D_C, t_L, $t(z)$ by adaptive `mpmath` quadrature at 25 significant digits, with the substitution $u = 1/(1+z)$ for the infinite bound of the age.
4. **Series expansions** and the closed form of the age verified by *computer algebra* (symbolic Sage), not only numerically.

B.2 Rebuilt parameters

Quantity	**SageMath (rebuilt)**	**astropy**
$\rho_{c,0}$	8.598814×10^{-30} g/cm^3	8.598814×10^{-30}
Ω_γ	5.4020151×10^{-5}	5.4020151×10^{-5}
$T_{\nu 0}$	1.9453688 K	1.9453688 K
Ω_ν (exact Fermi–Dirac)	1.4397093×10^{-3}	1.4396743×10^{-3} (Komatsu)
Ω_Λ (from flatness)	0.68884627	0.68884631
$\Omega_\mathrm{m} + \Omega_\nu$	0.31109971	0.31109968
t_0	13.786892 Gyr	13.786885 Gyr
$D_H = c/H_0$	14.4515552 Gly	14.4515552 Gly

The Komatsu fit departs from the exact integral by -0.0024 % at $z = 0$ and at most 0.12 % (around $z \sim 10$, in the relativistic/non-relativistic transition of the neutrinos). Since $\Omega_\nu \sim 10^{-3}$, the effect on the observables stays below 1.5×10^{-5}.

B.3 Agreement between the two computation chains

z	$E(z)$	D_C (Gly)	D_L (Gly)	D_A (Gly)	t_L (Gyr)	$t_{\rm em}$
0	1.0000	0.0000	0.000	0.00000	0.0000	13.7869 Gyr
0.00428	1.0020	0.0618	0.062	0.06153	0.0617	13.7252 Gyr
0.01	1.0047	0.1442	0.146	0.14275	0.1435	13.6434 Gyr
0.1	1.0502	1.4108	1.552	1.28258	1.3452	12.4417 Gyr
0.158	1.0826	2.1970	2.544	1.89726	2.0418	11.7451 Gyr
0.5	1.3188	6.3484	9.523	4.23224	5.1962	8.5906 Gyr
1	1.7829	11.0751	22.150	5.53754	7.9355	5.8513 Gyr
1.6	2.4819	15.2003	39.521	5.84625	9.7525	4.0344 Gyr
2.34	3.5056	18.8240	62.872	5.63592	10.9878	2.7991 Gyr
5	8.2452	25.9173	155.504	4.31956	12.6162	1.1707 Gyr
7.085	12.8620	28.8357	233.137	3.56657	13.0383	748.6 Myr
10.6	22.0837	31.8379	369.320	2.74465	13.3516	435.2 Myr
20	53.8259	35.7357	750.449	1.70170	13.6088	178.1 Myr
100	573.8216	41.8504	4 226.886	0.41436	13.7704	16.4 Myr
1 089.8	23 053	45.2848	49 396.658	0.04152	13.7865	372 kyr
1 500	38 877	45.4825	68 269.208	0.03030	13.7867	216 kyr

Quantity	**largest relative difference (16 values of z)**
$E(z)$	1.79×10^{-5}
D_C	2.10×10^{-6}
D_L	2.10×10^{-6}
D_A	2.10×10^{-6}
t_L	4.58×10^{-7}
$t_{\rm em}$	1.50×10^{-5}

Conclusion: the two calculations, independent from the choice of constants down to the quadrature method, agree far better than the uncertainty of the cosmological parameters themselves (~ 0.5 %, and ~ 7 % if one accounts for the Hubble tension). **The program is numerically reliable**; its limitation is the *model*, not the arithmetic.

B.4 Independent checks

- **Etherington $D_L = (1+z)^2 D_A$**: verified to 2×10^{-16} (machine precision) over the whole range.
- **Time conservation $t_L(z) + t_{\rm em}(z) = t_0$**: verified to 2×10^{-10} Gyr.
- **Closed form of the age** (18.1): recovered to 10^{-9} by comparison with the quadrature, for the model without radiation.
- **Series expansions** (13.2)–(13.5): the four coefficients $-\frac{1+q_0}{2}$, $+\frac{1-q_0}{2}$, $-\frac{3+q_0}{2}$, $-\frac{2+q_0}{2}$ are obtained by symbolic `series()` and confirmed numerically at $z = 0.1$ (D_C: 1.41076 against the exact 1.41077).

- **Maximum of D_A**: root of $(1+z)/E(z) = D_C(z)/D_H$, giving $z = 1.592133$ and $D_A = 5.84629$ Gly, identical by both methods to 4×10^{-7}.
- **Horizons**: particle horizon 46.2005 Gly, Hubble sphere 14.4516 Gly, event horizon 16.5808 Gly $= D_H \int_1^{\infty} \mathrm{d}a/[a^2 E(a)]$.
- **Correlation** $\rho(H_0, \Omega_\mathrm{m}) = -0.9763$, derived symbolically from the constraint on ω_m (section 6.5).
- **Curvature**: for $\Omega_k = -0.01$, 0, $+0.01$ at $z = 0.5$, 2.34, 10.6 and 1089.8, the values of D_C, D_M, D_L and D_A agree to **five decimals** between the program and SageMath.

B.5 The scripts

They ship with the program, in `verif_sage/`:

`verif_cosmo.sage` full reconstruction and reference table (runtime ~ 50 s);

`verif_DL_symbolique.sage` computer-algebra check of the series expansions and of the closed form of the age;

`verif_courbure_sigma.sage` symbolic derivation of the correlation $\rho(H_0, \Omega_\mathrm{m})$ and check of the distances with curvature.

Running them:

```
sage verif_sage/verif_cosmo.sage
```

Appendix C

Annotated bibliography and sources

All the numerical values in this course come from the Planck 2018 cosmology as implemented in `astropy.cosmology.Planck18`.

C.1 Measurement of the cosmological parameters

Planck 2018 (reference parameters). N. Aghanim *et al.* (Planck Collaboration), *Planck 2018 results. VI. Cosmological parameters*, Astron. Astrophys. **641**, A6 (2020). DOI | arXiv:1807.06209. The primary source of the parameters used (Table 2, column TT,TE,EE+lowE+lensing+BAO).

SH0ES (Hubble tension). A. G. Riess *et al.*, ApJL **934**, L7 (2022). arXiv:2112.04510.

Discovery of the acceleration. A. G. Riess *et al.*, AJ **116**, 1009 (1998); S. Perlmutter *et al.*, ApJ **517**, 565 (1999). Nobel Prize 2011.

C.2 Cosmological distance formulas

The practical reference. D. W. Hogg, *Distance measures in cosmology*, arXiv:astro-ph/9905116 (1999, updated 2000). Every distance formula in eleven pages; a daily-use document.

Reciprocity theorem. I. M. H. Etherington, *On the definition of distance in general relativity*, Phil. Mag. **15**, 761 (1933); reprinted in Gen. Rel. Grav. **39**, 1055 (2007).

Pedagogical approach. E. L. Wright, *A Cosmology Calculator for the World Wide Web*, PASP **118**, 1711 (2006). Describes the widely used calculator at astro.ucla.edu/∼wright/CosmoCalc.html.

C.3 Founding papers

Friedmann equations. A. Friedmann, *Uber die Krummung des Raumes*, Z. Phys. **10**, 377 (1922).

The Lemaitre–Hubble law. G. Lemaitre, Annales de la Societe Scientifique de Bruxelles **A47**, 49 (1927) — the first velocity–distance law, two years before Hubble. E. Hubble, PNAS **15**, 168 (1929) — the observational evidence.

FLRW metric. H. P. Robertson, ApJ **82**, 284 (1935); A. G. Walker, Proc. London Math. Soc. **42**, 90 (1937).

C.4 Textbooks

Introductory (upper undergraduate).

Ryden 2017 B. Ryden, *Introduction to Cosmology*, 2nd ed., Cambridge University Press. The standard starting point.

Liddle 2015 A. Liddle, *An Introduction to Modern Cosmology*, 3rd ed., Wiley. Very short and accessible.

Advanced (graduate).

Mukhanov 2005 V. Mukhanov, *Physical Foundations of Cosmology*, Cambridge.

Dodelson & Schmidt 2020 S. Dodelson and F. Schmidt, *Modern Cosmology*, 2nd ed., Academic Press. The current graduate reference for perturbations and the CMB.

Weinberg 2008 S. Weinberg, *Cosmology*, Oxford. Masterful.

Carroll 2004 S. M. Carroll, *Spacetime and Geometry*, Addison-Wesley. Chapter 8 = cosmology from general relativity.

C.5 Conceptual discussions

Origin of the cosmological redshift. E. F. Bunn and D. W. Hogg, *The kinematic origin of the cosmological redshift*, Am. J. Phys. **77**, 688 (2009).

Recession velocities. T. M. Davis and C. H. Lineweaver, *Expanding Confusion: Common Misconceptions of Cosmological Horizons and the Superluminal Expansion of the Universe*, PASA **21**, 97 (2004). The reference on superluminal velocities and horizons.

C.6 Online tools

- **NED** (NASA/IPAC Extragalactic Database) Cosmology Calculator
- **Ned Wright (UCLA) Cosmology Calculator**
- **ICRAR Cosmology Calculator**
- **Astropy** cosmology documentation

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